

Making Interactive Digital Artifacts for Learning

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QMUL-June-2026/](https://lpetrov.cc/AI-teaching/QMUL-June-2026/)

What's in this for me? A few links

- **Inspiration: Interactive probability materials, done "by hand":** [Seeing Theory](#), Brown, 2016–17
- **One of the first examples of my vibecoding:** [BBP transition / spiked random matrices](#)

Used in a random matrices course, Spring 2025.

Coded with web-based ChatGPT Pro at the time, with much frustration.

- **Research exploration:** [Aztec diamond with random weights](#)

Dimer models with random weights are a very active direction with few rigorous math results; looking at pictures is often all we have.

Coded with Claude Code (command-line coding agent from Anthropic).

- **PhD advising:** factorial Schur processes

See that my student Yizhen Li's computations in progress tie to probabilistic reality; put asymptotic analysis onto firm ground.

Coded with pi/codex (OpenAI model and open-source coding agent).

- **Student final project:** [bipartite dimers / lattice tool](#)

Done by undergraduate students at UVA Mathematical Experimentation Lab (I did not contribute to code).

- **My simulation collection:** lpetrov.cc/simulations

A lot of colleagues use these in talks and papers; a few pages include TikZ export.

The message

AI is now a very good tool for coding. For small standalone teaching and research artifacts, this lowers the barrier to entry to **zero — for everyone, not just people who already think of themselves as programmers.**

Before, the barrier was coding:

- How do I set up an HTML page?
- How do sliders, buttons, responsive animation, canvas/SVG work?
- How do I do simulations on the web fast?
- How do I make it look reasonable?
- How do I host it and share a link?
- Is this worth learning for one small demo?

Now the starting point can be:

- What should the student, researcher, or user manipulate?
- What should they predict, check, or see?
- What is the correct formula or edge case?
- What would make it useful for research discussion?

An AI-generated artifact is **not** automatically correct, accessible, or pedagogically good. But the first working version no longer requires a frontend-development project. The instructor's work moves from **implementation** to **design, judgment, checking, and revision**.

What's in this for you?

Some examples of where this might be useful for you — or not, which is fine.

Manipulatives for students

- One-off demos for concepts students need to manipulate, not just read about.
- Desmos is great for calculus.
- Probability beyond projects like Seeing Theory, and abstract algebra, often lack a simple visual equivalent.
- Sage is powerful, but not visual.

Instructor-side version: students do not have to use AI; they receive a standalone link, with no login, no accounts, no student-data collection.

Final/special projects

Students can create useful interactive artifacts as a final project, even if they have no coding background (but will have to use AI tools: check with your institution's policies).

How do I start?

Start with one bottleneck, not with “make me an app.”

Output: one standalone HTML/CSS/JavaScript page — interactive, no backend, no data transmitted.

- Chat apps can host these artifacts.
- Export to a course management system/LMS.
- Host on GitHub Pages, Google Sites, or any static website.
- Or just send the HTML file by e-mail.

Three hosted parabola explorers

- Claude version
- Gemini version
- ChatGPT Canvas version

Create a single-page interactive HTML/CSS/JavaScript learning artifact for university students.
Topic: two particles moving on a line.

Learning goal: students should understand relative velocity and the condition for two moving particles to meet.

Student task: particle 1 starts at position 0 and particle 2 starts at position d . Students should predict whether the particles meet, then press play to check the meeting time and position.

Include sliders for v_1 , v_2 , and d ; an animation; play/pause/reset buttons; axis labels; units; and two short guiding questions.

Correctness requirements: handle cases where the particles never meet, where relative velocity is zero, and where the meeting time would be negative.

Accessibility requirements: semantic HTML, keyboard-usable controls, visible focus, labels for all sliders/buttons, sufficient contrast, non-color cues, and a reduced-motion option.

Use no external libraries and do not collect or transmit data.

Output from this prompt: Claude: two-particle meeting simulator

What to be mindful of

- **Mathematical correctness:** formulas, assumptions, edge cases, units, terminology.
- **Pedagogy:** does the artifact create a real learning experience?
- **Accessibility:** keyboard use, visible focus, labels, contrast, non-color cues, motion control.
- **Student data:** pure HTML/JS manipulative with no backend and no external analytics is safe.
- **Hosting:** chat-app draft is fine for prototyping; course-ready versions may belong on GitHub Pages, course management systems, personal website, etc.
- **Check with your institution's policies** (which are inevitably in flux)

Context: I am a member of UVA's all-university AI Research Advisory Council (AIR Council). These are very initial findings.

Thank you

Slides and links:

lpetrov.cc/AI-teaching/QMUL-June-2026/



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